# Supervised Learning: Regression & Classification

Machine Learning â¢ Unit 1: Supervised Learning â¢ Study Notes

Loss functions, gradient descent optimization, and cross-entropy loss.

## 1. Empirical Risk Minimization

Hypothesis class h_theta(x) mapping input vectors to predictions.

Mean Squared Error (MSE) loss for regression: L(y, y_hat) = 1/2 * (y - y_hat)^2.

Binary cross-entropy loss for classification: -y log(p) - (1 - y) log(1 - p).

## 2. Gradient Descent Optimization

Parameter update: theta := theta - alpha * grad_theta(J(theta)).

Stochastic Gradient Descent (SGD) and mini-batch variance reduction.

L1 Lasso and L2 Ridge regularization for controlling overfitting.